

HARSH KASHYAP

 +91-8051625669  harsh.kashyap2001@gmail.com  [linkedin.com/in/harsh-kashyap](https://www.linkedin.com/in/harsh-kashyap)  github.com/Harsh23Kashyap  harshkashyapportfolio.netlify.app  leetcode.com/Harsh23Kashyap  [Certificates](#)

Skills

Languages & Backend: Java, Spring Boot, Python, C/C++, REST APIs, SQL
Web & UI: React.js, Tailwind CSS, JavaScript, HTML/CSS, Swift, SwiftUI, Flutter
Cloud & Infra: AWS (EC2, S3, SES, DynamoDB), Docker, Jenkins, CI/CD, Cloudflare
Data & AI: RAG, LLM agents, FAISS, XGBoost, TensorFlow/Keras, incident-resolution automation
Tools & Observability: Git/GitHub, Postman, Splunk, Shell scripting

Experience

Full Time

Apple | *Software Engineer 2* December 2023 – Present

- Built a **macOS Swift application** that replaced fragmented terminal-based 5G protocol testing workflows for **600+ engineers**, drove **80% adoption**, eliminated third-party tools, and saved **\$720K annually** while improving developer productivity and onboarding.
- Implemented a dynamic **pre-validation framework for LTE/NR message creation** that automatically handled interdependent component checks, reducing **debug time from 20 minutes to under 2** and improving incident **Mean-Time-to-Resolution (MTTR) by 40%** by proposing automated fixes based on previous fixes.
- Architected an **LLM-powered triage automation pipeline** using **RAG (FAISS-based embeddings)** with **CI/CD (Jenkins) integrations**, automating incident clustering and recommending fixes from historical logs, reducing engineering effort across **700+ incidents/month** for **60+ engineers** and scaling across **8 teams**.
- Led end-to-end bring-up of a cellular testing lab in India using an iPhone and **USRP X410/B210**, automating setup and testing with **Python** to reduce setup time from **3 weeks to under 2 days** and enable repeatable deployments.
- Led a greenfield initiative** to build LTE Layer 2 (RLC, PDCP, MAC) from scratch using a **multi-agent AI system**, orchestrating **56 agents** with **5 domain experts** to generate and validate **100K+ lines of C/C++ code**; designed a 3GPP-aware hybrid retrieval system (BM25 + dense + RRF fusion) to ground outputs in telecom specifications, using UE simulation and human-in-the-loop validation to reduce development cycles from ~3.5 years to ~6 months.

PayPal | *Software Engineer* July 2023 – October 2023

- Improved the Global Merchant Lending admin tool, driving **13% audit efficiency gain** by automating manual workflows using **React, Node.js**, and internal PayPal frameworks; ensured **CECL** compliance.

Internships

Apple | *Software Developer Engineering Intern* January 2023 – June 2023

- Created an internal single-page application (SPA) using **React.js, Tailwind CSS, Spring Boot, Splunk, and internal frameworks**, including **real-time observability dashboards** that automated recurring health checks and led to an **83% reduction in support turnaround times**.
- Designed and deployed an ML pipeline (**XGBoost + feature engineering**) to predict SLA resolution times with **80%+ accuracy**, reducing SLA breaches by **25%** across IS&T and ticket incidents.

Tata 1mg | *Engineering Intern* October 2022 – November 2022

- Utilized **Asyncio, Celery, and RabbitMQ** for asynchronous task scheduling, delivering **operational efficiencies for 3,000 employees** in mail scheduling and projecting potential benefits for **over 4 million customers**.

Cisco | *Technical Undergraduate Intern* June 2022 – July 2022

- Conducted SDWAN network data analysis, created an internal dashboard displaying Quality of Service metrics, and achieved a **23.2% efficiency improvement using Data Science, Python, and networking tools**.

Licious | *Software Developer Intern* January 2022 – May 2022

- Successfully migrated sections of the application from **Swift to Flutter**, involving extensive **refactoring of over 5,000 lines of code** and resulting in an **estimated 30% reduction in man-hours** and **improved app stability**.

YASH Technologies | *Software Engineering Intern* July 2021 – September 2021

- Collaborated on the development of an Employee Management System portal using **Java, Spring Boot, and SQL**, successfully integrating a database for attendance tracking of **over 6,000 employees**.

Research

Professor Dennis Shasha | *New York University*

- Developed **CustomNerd**, a modular framework at **New York University (NYU)**, under submission as an MPDI research paper and adopted across institutions, enabling new models in **under 5 hours**; powering systems like **DietNerd** (PubMed), **WirelessNerd** (3GPP), and **InvestorNerd** (Yahoo Finance), with safe LLM design on **AWS (EC2/S3), Cloudflare**, load balancing, and OTP-based auth via **DynamoDB** and **SES**.

Dr. Parteek Kumar | *Washington State University*

- Prototyped a dual-agent job matching system using **FastAPI, SentenceTransformers, and Chroma/Qdrant** for semantic search, enabling configurable ranking and explainable scores via a **React dashboard**.

Education

B.E., Computer Science	2019 – 2023	Thapar Institute of Engineering and Technology	9.71 CGPA
Class XII (ISC)	2019	Mount Assisi School	96.25%
Class X (ICSE)	2017	Mount Assisi School	94.33%

Achievements

- Awarded the **TIET Merit Scholarship** for ranking in the **top 0.5% all four years** of academic studies.
- Ranked **89th out of 100,000** students in the IICC by AICTE coding competition.
- Featured in Apple's Innovation Deck for an idea on Apple Pay, selected out of 350+ submissions.